



**School of  
Engineering**

IDP Institut für Datenanalyse  
und Prozessdesign



## Fall Design Report

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UAV Health Monitoring

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Submission Date: November 25<sup>th</sup>, 2025

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## The Team

### Team Members

- Prishha Chawla [CS]
- Siddharth Urankar [CS]
- Ally Blair [ME]

### Team Advisors

- Dr. Manish Kumar [UC ME]
- Dr. Manuel Arias Chao [ZHAW, Tu Delft]
- Dr. Chetan Kulkarni [NASA]
- Luke Busse [PhD Student at UC]

## Project Abstract

This work presents a UAV powertrain health-monitoring framework that integrates a semi-supervised CNN autoencoder for early detection of motor acoustic anomalies with a supervised DeepONet that models battery degradation, predicts remaining useful life, and estimates flight time. Together, these models deliver real-time diagnostics to enhance reliability and safety in mission-critical operations.

## Project Description

The aim of this project is to develop a UAV powertrain health monitoring framework that integrates hardware based acoustic sensing with machine learning driven diagnostics. The system will address two UAV subsystems: motors and batteries, which are among the most failure prone components in UAV operations.

The project involves building a UAV test platform equipped with USB microphones, current/voltage sensors, and a drone stand for controlled, repeatable, and safe testing. Microphone placement has been selected to balance signal to noise ratio so that both motor and propeller related faults can be captured. Data will be recorded in two ways: in a stand and while in flight to validate motor performance under realistic conditions. Multiple methods of recording motor and propeller sound will be designed and compared, along with a strain measuring system that will advise the health monitoring framework regarding structural stresses. Two models will be developed and integrated: a 1D CNN Autoencoder for semi supervised detection of motor and propeller anomalies from acoustic data, and a DeepONet for supervised battery degradation modeling, RUL prediction, and flight time estimation. Both models will be optimized for deployment on onboard computer hardware with a NVIDIA Jetson Nano used for in-flight computing.

The final system will offer anomaly detection and diagnostics while in flight. This project and the associated research directly enables integration into the IASMS by combining deep learning, mechanical design, and hardware sensing. This will enhance UAV safety and reliability for all applications.

## User Stories and Design Diagrams

User stories portray some possible use cases of this work from a user's perspective. The design diagrams start with level 0 [being the most basic and simple version] and go on to levels 2 and 3, that are increasingly detailed about the design of this work.

### User Stories

1. As a UAV operator, I want to automatically detect motor or propeller faults during flight, so that I can ensure safe landing and prevent crashes in mission-critical operations.
2. As a drone fleet manager, I want to track battery health and predict remaining flight time, so that I can plan missions more reliably and reduce unexpected downtime.
3. As a remote operator using a drone-in-a-box system, I want in-flight monitoring of propeller damage, so that I can identify issues that are not visually observable and avoid risky deployments.
4. As a UAV maintenance technician, I want pre-flight diagnostics to flag potential motor or propeller issues, so that I can repair or replace components before they fail in operation.
5. As an aviation safety regulator, I want a standardized health monitoring framework integrated into UAVs, so that I can improve operational safety and ensure compliance with safety management systems.

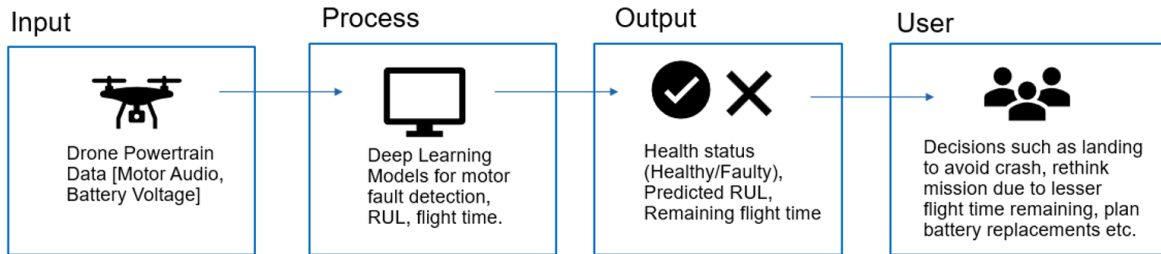
### Design Diagrams

#### *Conventions*

- Boxes represent a set of information, labelled where necessary such as input, output etc.
- Arrows represent transition from one box to the next, i.e., from one set of steps to the next.
- 2-sided arrows represent an extension of information provided by 1 box, in another box.
- Blue boxes and arrows represent the highest-level view of the design.
- Additional yellow boxes and arrows represent a lower-level view.
- Green boxes and arrows represent the lowest level view of the design.

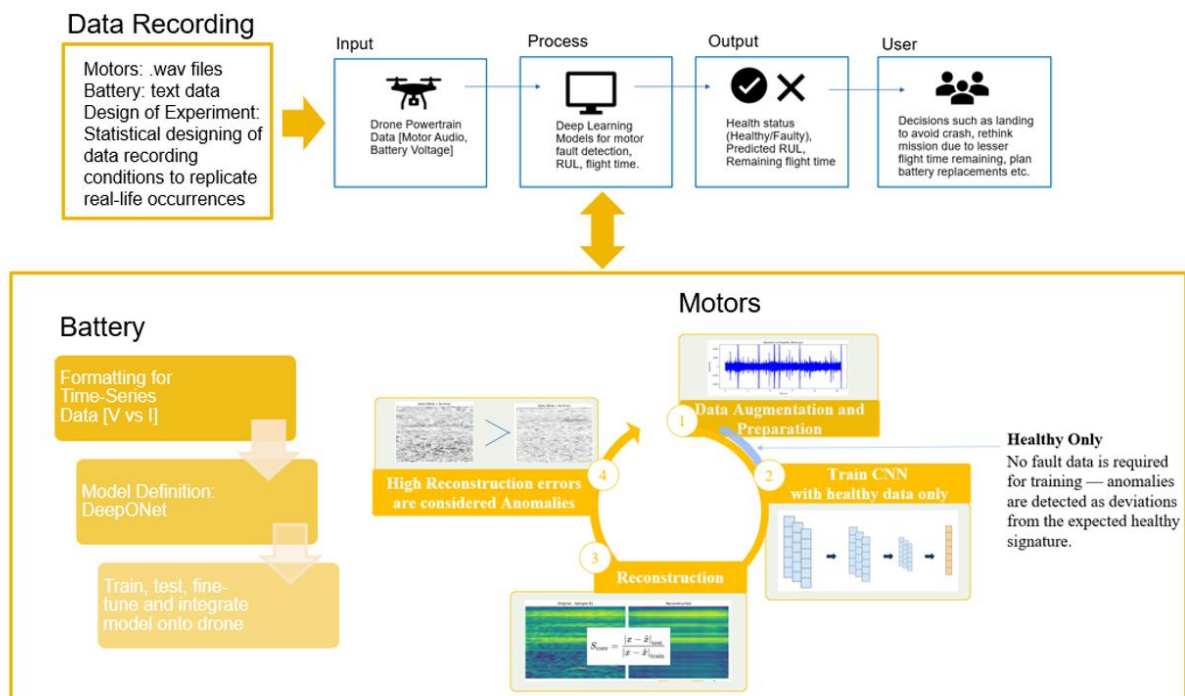
### Design Level 0

The following flowchart portrays this project in a simple input-output format using blue boxes and arrows.



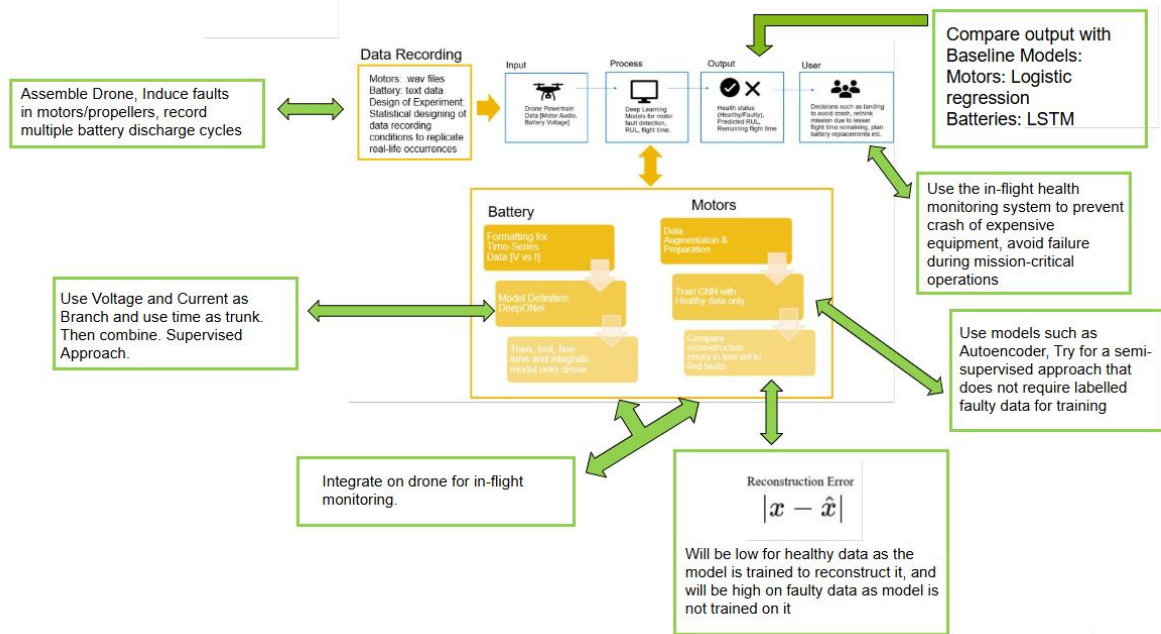
### Design Level 1

This deeper design diagram builds upon the level 0 flow chart and adds detail using yellow boxes and arrows.



### Design Level 2

Building further on the diagram from level 1, level 2 adds another layer of detail using green arrows and boxes.



## Project Tasks and Timeline

### Task List

The tasks are grouped by the team member responsible for the task and each task is further broken down with a suitable description.

#### *Ally Blair (ME) – Mechanical Systems, Data Collection*

1. **Assemble UAV and test platform** with stable mounting for motors and sensors.
2. **Conduct vibration isolation experiments** to reduce structural noise in audio recordings.
3. **Compare microphone types (MEMS vs USB)** under identical test conditions.
4. **Determine optimal microphone placement** through controlled bench testing.
5. **Induce controlled propeller faults** (e.g., chipped, unbalanced, bent blades) for dataset generation.
6. **Induce controlled motor faults** (e.g., bearing wear, partial short) and record signatures.
7. **Establish safety protocols** for running UAVs with faulty motors during testing.
8. **Record baseline healthy motor datasets** across different operating conditions (RPM, load, environment).
9. **Record faulty motor datasets** at varying fault severity levels for training/evaluation.
10. **Document mechanical test setup** including fault induction procedures, mic placement diagrams, and UAV configuration.

*Prissha Chawla (CS) – Data Science, AI, ML*

11. **Design of experiment plan** detailing recording scenarios, environmental variations, and repetitions.
12. **Perform exploratory data analysis (EDA)** on collected motor audio datasets.
13. **Convert audio data into spectrograms** and evaluate preprocessing pipelines.
14. **Develop semi-supervised autoencoder model** for anomaly detection on healthy motor audio.
15. **Evaluate reconstruction error distributions** to establish anomaly detection thresholds.
16. **Investigate supervised ML approaches** (e.g., DeepONet or alternatives) for UAV battery health monitoring.
17. **Train and validate ML models** using induced fault datasets and battery profiles.
18. **Conduct hyperparameter tuning** for both autoencoder and supervised models.
19. **Benchmark model performance** (precision, recall, F1) against UAV health monitoring objectives.
20. **Document AI methodology and results** for inclusion in project reports and publications.

*Siddharth Urankar (CS) – Software, Integration, Deployment*

21. **Install and configure Linux OS** on NVIDIA Jetson for UAV onboard computing.
22. **Develop microphone interface software** to support synchronized acoustic data capture.
23. **Implement real-time data logging system** for motor audio during UAV operation.
24. **Optimize data storage format** for efficiency in transfer and later ML model use.
25. **Deploy trained autoencoder model** onto Jetson for onboard inference.
26. **Develop anomaly detection pipeline** that integrates audio capture with autoencoder output.
27. **Test latency and throughput** of model inference on Jetson under flight-like conditions.
28. **Implement battery data collection interface** for supervised RUL prediction model.
29. **Optimize Jetson codebase** to balance recording, inference, and data transfer simultaneously.
30. **Validate end-to-end system integration** by running real-time UAV health monitoring tests.

## Milestones

| #  | Milestone                               | Description                                                                                                           | Expected Completion |
|----|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|
| 1  | Motor and Propeller Data Recording      | Collection of baseline and faulty motor/propeller datasets under controlled lab conditions to support model training. | 8 Oct               |
| 2  | Initial Motor Model Developed           | Development of first version of the motor health model using recorded data to detect motor faults and anomalies.      | 22 Oct              |
| 3  | Integration on Jetson Hardware          | Integrate initial motor model with onboard Jetson computer for in-flight real-time inference.                         | 29 Oct              |
| 4  | Battery Data Round 1 Collected          | Perform first round of UAV battery discharge tests and record related performance metrics.                            | 24 Oct              |
| 5  | In-flight Motor Testing Conducted       | Deploy and validate motor anomaly detection model in real flight-like environments.                                   | 5 Nov               |
| 6  | Motor Model Fine-tuned                  | Improve accuracy and reduce false positives in motor fault detection through retraining.                              | 19 Nov              |
| 7  | Battery Model Prototyped                | Develop an initial model for predicting battery health and remaining life using recorded data.                        | 28 Nov              |
| 8  | Strain Gauges & Second Drone Integrated | Implement strain gauges and secondary microphone array on second UAV for extended testing.                            | 3 Dec               |
| 9  | Extended Battery Data Collected         | Conduct additional battery experiments to capture degradation trends under varying conditions.                        | 31 Dec              |
| 10 | Motor System Documentation Finalized    | Prepare detailed documentation of test setup, model performance, and data pipeline.                                   | 31 Dec              |
| 11 | Battery Model Fine-tuned and Validated  | Finalize battery model and integrate results into UAV software for full system operation.                             | 20 Jan              |
| 12 | First Design Review Completed           | Present first major checkpoint on progress, design, and model performance.                                            | 1 Dec               |



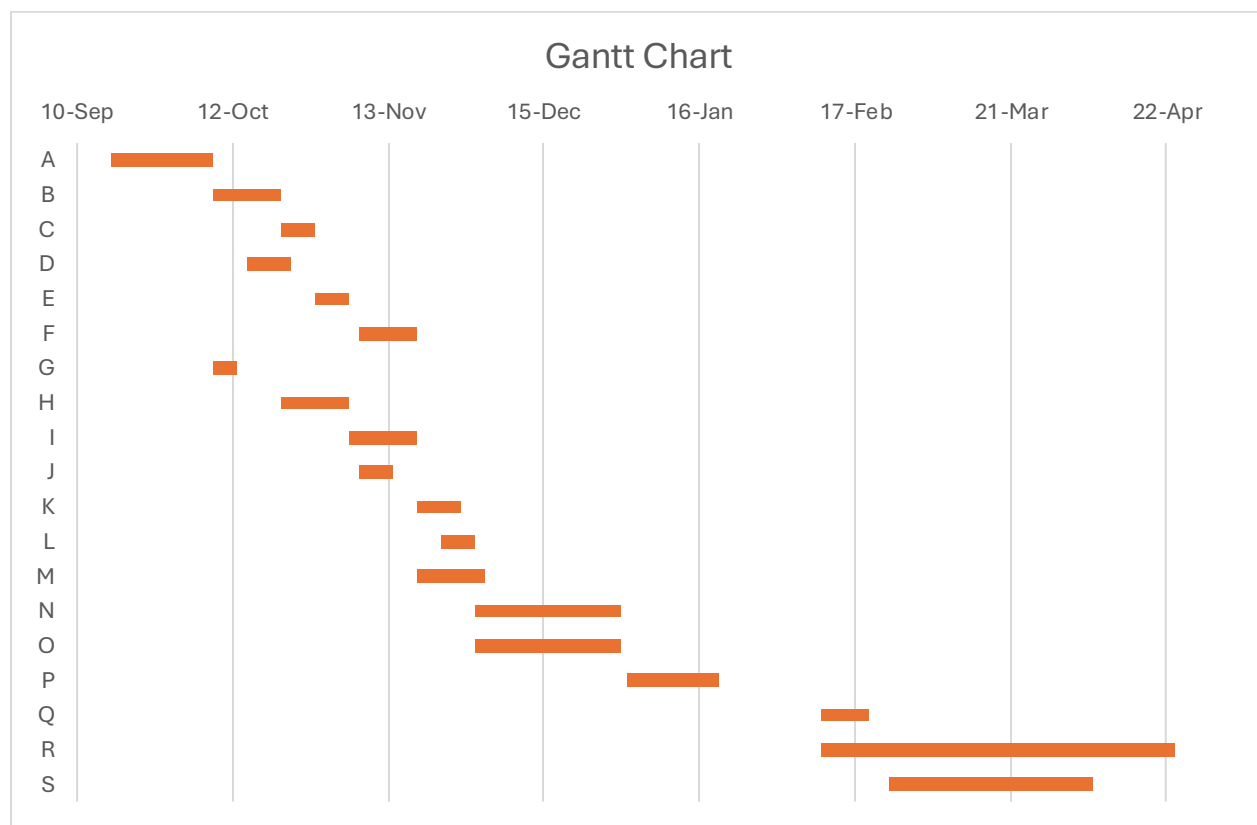
|           |                                             |                                                                                              |        |
|-----------|---------------------------------------------|----------------------------------------------------------------------------------------------|--------|
| <b>13</b> | Second Design Review Completed              | Present full system results and progress updates before final evaluation.                    | 20 Feb |
| <b>14</b> | Final Report and Expo Presentation Prepared | Compile final report and prepare for Expo demonstration of UAV health monitoring system.     | 7 Apr  |
| <b>15</b> | Final Report Submission                     | Submit detailed project report summarizing all development, testing, and evaluation results. | 24 Apr |

## Timeline

| <b>Task ID</b> | <b>Task Description</b>                    | <b>Start Date</b> | <b>End Date</b> | <b>Duration (days)</b> | <b>Milestone</b> |
|----------------|--------------------------------------------|-------------------|-----------------|------------------------|------------------|
| <b>A</b>       | Record motor and propeller data            | 17-Sep            | 8-Oct           | 21                     | M1               |
| <b>B</b>       | Create initial motor model                 | 8-Oct             | 22-Oct          | 14                     | M2               |
| <b>C</b>       | Integrate motor model with Jetson          | 22-Oct            | 29-Oct          | 7                      | M3               |
| <b>D</b>       | Record round 1 battery data                | 15-Oct            | 24-Oct          | 9                      | M4               |
| <b>E</b>       | In-flight motor testing                    | 29-Oct            | 5-Nov           | 7                      | M5               |
| <b>F</b>       | Fine-tune motor model                      | 7-Nov             | 19-Nov          | 12                     | M6               |
| <b>G</b>       | Decision-making on battery discharge plan  | 8-Oct             | 13-Oct          | 5                      | M4               |
| <b>H</b>       | Build second drone                         | 22-Oct            | 5-Nov           | 14                     | M8               |
| <b>I</b>       | Integrate and test second microphone array | 5-Nov             | 19-Nov          | 14                     | M8               |
| <b>J</b>       | Round 2 battery data collection            | 7-Nov             | 14-Nov          | 7                      | M7               |
| <b>K</b>       | Create initial battery model               | 19-Nov            | 28-Nov          | 9                      | M7               |
| <b>L</b>       | Prepare for 1st Design Review              | 24-Nov            | 1-Dec           | 7                      | M12              |
| <b>M</b>       | Integrate strain gauges on second drone    | 19-Nov            | 3-Dec           | 14                     | M8               |
| <b>N</b>       | Further battery data collection            | 1-Dec             | 31-Dec          | 30                     | M9               |

|          |                                    |        |        |    |     |
|----------|------------------------------------|--------|--------|----|-----|
| <b>O</b> | Motor documentation                | 1-Dec  | 31-Dec | 30 | M10 |
| <b>P</b> | Test and fine-tune battery model   | 1-Jan  | 20-Jan | 19 | M11 |
| <b>Q</b> | Prepare for 2nd Design Review      | 10-Feb | 20-Feb | 10 | M13 |
| <b>R</b> | Write final report                 | 10-Feb | 24-Apr | 73 | M15 |
| <b>S</b> | Expo preparation and participation | 24-Feb | 7-Apr  | 42 | M14 |

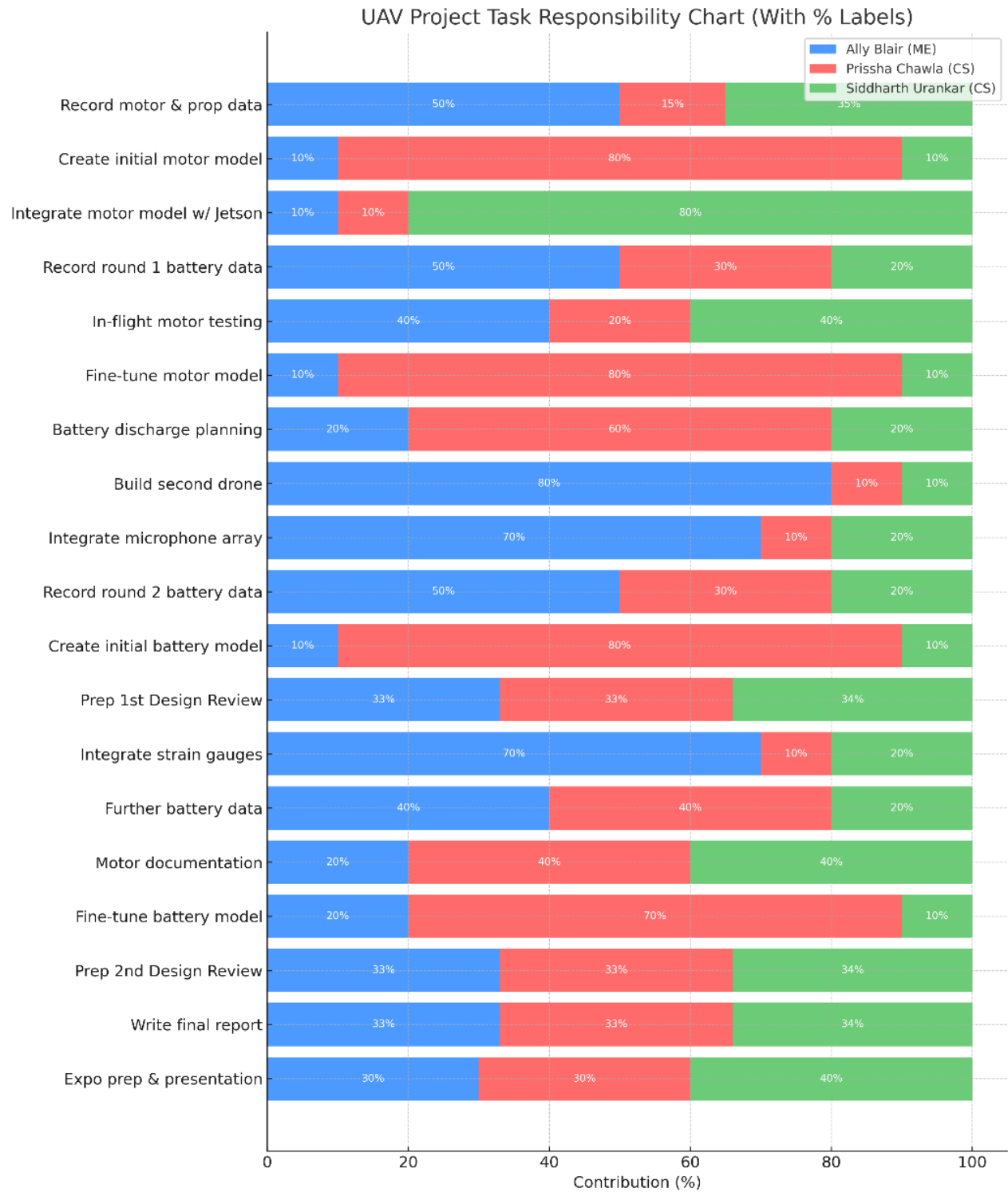
### Timeline Gantt Chart



## Effort Matrix

| <b>Task ID</b> | <b>Task Description</b>                 | <b>Ally Blair (%)</b> | <b>Prissha Chawla (%)</b> | <b>Siddharth Urankar (%)</b> | <b>Primary Responsibility</b> |
|----------------|-----------------------------------------|-----------------------|---------------------------|------------------------------|-------------------------------|
| <b>A</b>       | Record motor and propeller data         | 50                    | 15                        | 35                           | Ally Blair                    |
| <b>B</b>       | Create initial motor model              | 10                    | 80                        | 10                           | Prissha Chawla                |
| <b>C</b>       | Integrate motor model with Jetson       | 10                    | 10                        | 80                           | Siddharth Urankar             |
| <b>D</b>       | Record round 1 battery data             | 50                    | 30                        | 20                           | Ally Blair                    |
| <b>E</b>       | In-flight motor testing                 | 40                    | 20                        | 40                           | Siddharth Urankar             |
| <b>F</b>       | Fine-tune motor model                   | 10                    | 80                        | 10                           | Prissha Chawla                |
| <b>G</b>       | Battery discharge planning              | 20                    | 60                        | 20                           | Prissha Chawla                |
| <b>H</b>       | Build second drone                      | 80                    | 10                        | 10                           | Ally Blair                    |
| <b>I</b>       | Integrate second microphone array       | 70                    | 10                        | 20                           | Ally Blair                    |
| <b>J</b>       | Record round 2 battery data             | 50                    | 30                        | 20                           | Ally Blair                    |
| <b>K</b>       | Create initial battery model            | 10                    | 80                        | 10                           | Prissha Chawla                |
| <b>L</b>       | Prepare for 1st Design Review           | 33                    | 33                        | 34                           | All                           |
| <b>M</b>       | Integrate strain gauges on second drone | 70                    | 10                        | 20                           | Ally Blair                    |
| <b>N</b>       | Further battery data collection         | 40                    | 40                        | 20                           | Ally Blair                    |
| <b>O</b>       | Motor documentation                     | 20                    | 40                        | 40                           | Siddharth Urankar             |
| <b>P</b>       | Fine-tune battery model                 | 20                    | 70                        | 10                           | Prissha Chawla                |
| <b>Q</b>       | Prepare for 2nd Design Review           | 33                    | 33                        | 34                           | All                           |
| <b>R</b>       | Write final report                      | 33                    | 33                        | 34                           | All                           |
| <b>S</b>       | Expo preparation and presentation       | 30                    | 30                        | 40                           | Siddharth Urankar             |

## Effort Chart



## ABET Concerns

The development of an in-flight UAV powertrain health monitoring framework is shaped by several important constraints that influence both technical and non-technical design decisions. These constraints affect hardware selection, machine learning approach, testing procedures, and the overall scope of what can be achieved in the project. Addressing them carefully ensures that the final system remains feasible, safe, and aligned with real-world operational conditions.

Economic constraints are significant because this is an undergraduate project with limited funding. High-end microphones and specialized sensors such as those used at NASA by Dr. Kulkarni are not feasible to procure. Instead, cost-effective microphones and open-source software are being used, which impacts accuracy, scalability, and sensor fidelity.

Professional constraints also play a role because the project involves advanced mechanical engineering, sensor integration, and machine learning. This requires careful planning, disciplined execution, and maintaining engineering standards that reflect well on all collaborators and the institutions involved.

Ethical considerations are critical because the system is designed to detect faults in aerial vehicles. In real-world operations, detecting a healthy system as faulty is preferable to missing a fault and allowing a UAV to fly in unsafe conditions. Faulty real-world data is limited, and supervised models would likely bias toward healthy data due to class imbalance. A semi-supervised approach is therefore used to build a more reliable system that prioritizes safety over raw performance.

Legal constraints are relevant because UAV testing must comply with federal and institutional safety regulations. This limits where and when flight testing can occur and imposes strict requirements on flight procedures. Security is not a major technical constraint since the collected data will be made public, but UAVs are often deployed in security-sensitive applications. This raises the importance of safe operation in environments where drones may face interference or hazardous conditions.

Environmental and social factors influence the project through its expected benefits. By enabling early fault detection and predictive maintenance, the project aims to reduce crash risks, minimize environmental debris, and support more reliable UAV deployment in applications such as inspection, logistics, and emergency response. Diversity and cultural impact are represented through the interdisciplinary structure of the team. Meeting both mechanical engineering and computer science requirements adds coordination complexity, but also strengthens the project by combining different areas of expertise.

These constraints collectively shape the system's design choices and research direction. By recognizing them early and building the project around realistic technical and operational boundaries, the team can deliver a solution that is safe, feasible, and impactful in real-world UAV applications.

## Self-Assessment

My dual major in Computer Science and Statistics has prepared me to contribute to this project at the intersection of algorithms, modeling, and real-world data. The overlap of these two fields, statistical rigor in designing experiments and evaluating models combined with computational power in implementing deep learning architectures, defines my role on this project. Specifically, I focus on developing the models that will diagnose UAV motor health from acoustic data and predict Li-ion battery degradation through voltage discharge curves. My statistical training ensures I can design and validate these models with confidence, while my computer science background allows me to build robust and scalable deep learning pipelines. In addition, a workshop in Taiwan gave me hands-on experience building and programming a robot, which introduced me to hardware integration and strengthened my ability to collaborate across disciplines. Even though my primary role is modeling and data analysis, this exposure allows me to better understand the mechanical aspects of drones and contribute meaningfully when hardware and software meet.

My three co-op rotations at Delta Air Lines played a critical role in shaping both my technical and professional preparation for this project. Over two years, I worked as an HR Analytics and Insights Co-Op, where I built skills in data wrangling, visualization, predictive modeling, and reporting automation. While the focus was on HR datasets, the experience gave me a strong foundation in handling large-scale, real-world data and taught me the value of scalability and accuracy. Just as importantly, working at a major airline sparked my interest in aviation and aerospace as industries where safety and reliability are non-negotiable. I learned how to analyze data with precision, communicate findings effectively, and build dashboards that told a clear story to decision-makers. These lessons extend directly to UAV health monitoring, where clarity and dependability are just as vital. At Delta, I also saw firsthand how data-driven systems could support mission-critical operations, and that perspective continues to guide my approach to this project.

My international exchange semester at the Zurich University of Applied Sciences (ZHAW) further deepened my connection to UAV safety research. While there, I enrolled in a data science course project under Professor Manuel Arias Chao in collaboration with Dr. Chetan Kulkarni from NASA Ames. Together, they introduced a project on UAV motor diagnostics using acoustic data, where I conducted a feasibility study with small online datasets. This work successfully demonstrated the potential of audio-based anomaly detection and sparked further interest in the topic. After returning to UC, I joined Dr. Manish Kumar's CS4005 Special Topics in Computer Science course, where I collaborated with Dr. Chao and Dr. Kulkarni again to conduct a second feasibility study, this time on UAV battery health monitoring. Passing both the motor and battery feasibility tests provided a strong foundation for scaling the work into a full senior design project. With the support of Dr. Kumar, his PhD student Luke, and my international collaborators, I formed a team to take this project forward into real-world UAV data collection and advanced modeling.

I am motivated to continue this project because it represents the culmination of multiple years of academic work, international research experiences, and interdisciplinary collaboration. My interest in aviation and aerospace was strengthened by my co-op experiences at Delta Air Lines, but it was my research with Dr. Chao, Dr. Kulkarni, and Dr. Kumar that showed me the exciting possibilities of combining machine learning with aerospace safety. This project is meaningful to me because it is not only technically challenging but also impactful in ensuring UAV reliability in mission-critical settings. I am excited to take the feasibility studies I completed abroad and at UC into a real-world implementation with my senior design team. The opportunity to work under leading faculty and NASA researchers provides a unique learning experience while also allowing me to contribute to safety solutions with real-world applications. The interdisciplinary nature of the project, spanning data science, mechanical systems, and aerospace engineering, makes it both challenging and inspiring.

My preliminary approach to this project is structured around scaling from controlled feasibility tests to full UAV deployment. Working with my team, I will help design the data collection process for both motor acoustics and battery performance during drone flights. Once data is collected, I will focus on preprocessing methods such as spectrogram generation and normalization, followed by model development. For motors, I will implement semi-supervised autoencoders to detect deviations from healthy acoustic patterns, while for batteries, I will train supervised models such as DeepONet to predict degradation and Remaining Useful Life. The expected outcomes are reliable fault detection and accurate flight time predictions, which will together form an integrated UAV safety monitoring framework. I will evaluate my contributions by measuring model accuracy and robustness, ensuring reproducibility, and confirming that the models integrate seamlessly into our broader system. I will know I have succeeded when the framework not only works on our collected data but also demonstrates value to our advisors and collaborators in the aerospace field.

## Professional Biography

### Prissha Chawla – Computer Science and Statistics

- Student at the University of Cincinnati, B.S. Computer Science and Statistics, Certificates in Analytics and Cloud Computing, Expected Graduation: Spring 2026
- International experience: Exchange at ZHAW, Switzerland; NTUT PBL Robotics Workshop, Taiwan
- Co-op work experience: HR Analytics & Insights Co-Op, Delta Air Lines; Software Developer Co-Op, Procter & Gamble; Cloud Engineering Trainee, Hitachi Systems
- Skills: Python, R, SQL, deep learning frameworks (CNN, LSTM, DeepONet), predictive modeling, data wrangling, visualization, cloud services

- Project role: Development of 1D CNN autoencoder for motor anomaly detection, battery modeling, experiment design, exploratory data analysis, AI methodology documentation

### Siddharth Urankar – Computer Science

- Student at the University of Cincinnati, B.S. Computer Science, Minor in Mathematics, Certificates in Software Engineering and Business Analytics, Expected Graduation: Spring 2026
- Studied abroad at ZHAW School of Engineering, Switzerland; participating in Ruhr Fellowship, Germany
- Interests: Artificial intelligence, software engineering, applications at the intersection of energy and financial systems
- Project role: Jetson integration, battery model development, software deployment
- Relevant experience: Python, PyTorch, computer vision, analytics tools, DeepONet and related pipelines for battery health monitoring and remaining flight time estimation

### Ally Blair – Mechanical Engineering

- Student at the University of Cincinnati, B.S. Mechanical Engineering, Expected Graduation: Spring 2026
- Relevant experience: UAV assembly, sensor integration, vibration isolation, test stand design, electromechanical systems
- Skills: SolidWorks, Siemens NX, Autodesk Inventor, mechanical design, system integration, custom fixture and test system design
- Project role: Construction of UAV platforms, integration of microphones, strain gauges, and electrical sensors, mechanical testing and validation
- Other experience: Hands-on laboratory projects, interdisciplinary collaboration with software and AI teams

### Budget

This project does not have a designated budget and as and when need arises for equipment, requests are filled in for approval. This project is being conducted at the UC's Applied Autonomy Lab and all donations are from other projects being conducted at this lab under our advisor, Dr. Manish Kumar. Exact value of donated items is not known and it is harder to determine due to the nature of this project using damaged equipment [for anomaly detection].



## Material Cost Breakdown

| Item                                                  | Estimated Cost (USD) | Notes                      |
|-------------------------------------------------------|----------------------|----------------------------|
| USB microphones                                       | 22                   | Purchased                  |
| MEMS I2S microphones                                  | 12                   | Purchased                  |
| Breadboards and associated wires                      | 10                   | Purchased                  |
| Strain gauges                                         | 40                   | Purchased                  |
| Thermocouple / Thermistor                             | 40                   | Purchased                  |
| Additional sensors                                    | TBD                  | To be acquired as needed   |
| Lightly used, new, damaged propellers and BLDC motors | N/A                  | Donated from lab inventory |
| Components to build second UAV                        | N/A                  | Donated from lab inventory |
| NVIDIA Jetson Nano                                    | N/A                  | Donated from lab inventory |
| <b>Total New Material Costs</b>                       | 124                  | Excludes donated items     |

## Appendix

### Project Hours

These hours can be verified by our advisor Luke Busse who is in the lab on Wednesdays and Fridays, and other meetings can be justified by meeting notes and the faculty advisors themselves. ME Senior design section attendance can be verified by Dr. Jacob Cress. The hours put in by team members working remotely on their specified tasks and doing research or preparing for lab working sessions are not included here.

| Team Member              | Activity Description                 | Frequency/Duration            | Total Hours |
|--------------------------|--------------------------------------|-------------------------------|-------------|
| <b>Siddharth Urankar</b> | DF Lab work with team                | 12 Wednesdays $\times$ 3 hr   | 36          |
|                          | CS Senior Design Section Assignments | 0.5 hr/week $\times$ 14 weeks | 7           |
|                          | Friday project work                  | 3 Fridays $\times$ 3 hr       | 9           |
| <b>Total</b>             |                                      |                               | <b>52</b>   |

|                       |                                      |                                    |             |
|-----------------------|--------------------------------------|------------------------------------|-------------|
| <b>Prishha Chawla</b> | DF Lab work with team                | 12 Wednesdays $\times$ 3 hr        | 36          |
|                       | ME Senior Design Section             | 8 Wednesdays $\times$ 2 hr         | 16          |
|                       | Dr. Chao weekly meeting              | 12 Wednesdays $\times$ 1 hr        | 12          |
|                       | Dr. Kulkarni monthly meeting         | $\sim$ 3 hr                        | 3           |
|                       | Friday project work                  | 3 Fridays $\times$ 3 hr            | 9           |
|                       | CS Senior Design Section Assignments | 0.5 hr/week $\times$ 14 weeks      | 7           |
|                       | ME Senior Design Section Assignments | 0.25 hr/week $\times$ 14 weeks     | 3.5         |
| <b>Total</b>          |                                      |                                    | <b>86.5</b> |
| <b>Ally Blair</b>     | DF Lab work with team                | 12 Wednesdays $\times$ 3 hr        | 36          |
|                       | ME Senior Design Section             | 12 Wednesdays $\times$ 2 hr        | 24          |
|                       | Check-in with Luke/Dr. Kumar         | $\sim$ 1 hr/week $\times$ 10 weeks | 10          |
|                       | Friday project work                  | 3 Fridays $\times$ 3 hr            | 9           |
|                       | ME Senior Design Section Assignments | 0.5 hr/week $\times$ 14 weeks      | 7           |
| <b>Total</b>          |                                      |                                    | <b>86</b>   |

## Links

Link to GitHub: [Link](#)

Link to OneDrive: [Senior Design 1 - OneDrive](#)

## References:

- [1] J. Novotňák, M. Fil'ko, P. Lipovský and M. Šmelko, "Design of the System for Measuring UAV Parameters," *Drones*, vol. 6, no. 8, p. 17, 2022.
- [2] . Y. Kato and M. Otaka, "Compressed Sensing of Vibration Signal for Fault Diagnosis of Bearings, Gears, and Propellers Under Speed Variation Conditions," *Sensors*, vol. 25, no. 10, p. 24, 2025.

- [3] V. Alecsandri and S. cel Mare, "Finding fault types of BLDC motors within UAVs using machine," *Heliyon*, 2024.
- [4] T. Zhao, Y. Zhang, M. Wang, W. Feng, S. Cao and G. Wang, "A Critical Review on the Battery System Reliability of Drone Systems," *Drones*, vol. 9, no. 8, p. 59, 2025.
- [5] A. Theissler, J. Pérez-Velázquez and M. Kettelgerdes, "Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry," *Reliability Engineering & System Safety*, vol. 215, 2021.
- [6] X. Wang, D. Mao and X. Li, "Bearing fault diagnosis based on vibro-acoustic data fusion and 1D-CNN network," *Measurement*, vol. 173, 2021.
- [7] S. Chen, J. Yu and S. Wang, "One-dimensional convolutional auto-encoder-based feature learning for fault diagnosis of multivariate processes," *Journal of Process Control*, vol. 87, 2018.
- [8] M. J. Bianco, P. Gerstoft, J. Traer, E. R. Ozanich, M. A. Roch, S. Gannot, C. Deledalle and W. Li, "Machine learning in acoustics: Theory and applications," *Acoustical Society of America*, 2019.
- [9] G. Lannace, G. Ciaburro and A. Trematerra, "Acoustical unmanned aerial vehicle detection in indoor scenarios using logistic regression model," *Applied Acoustics*, vol. 28, no. 1, 2020.
- [10] F. Xiao, Y. Ding, X. Li, K. Zhang and J. Liu, "A CNN-LSTM Method Based on Voltage Deviation for Predicting the State of Health of Lithium-Ion Batteries," *Bioengineering & Translational Engineering*, vol. 4, no. 3, 2024.
- [11] Y. Huang, X. Gong, Z. Lin and L. Xu, "A hybrid data-driven method for voltage state prediction and fault warning of Li-ion batteries," *Case Studies in Thermal Engineering*, vol. 64, 2024.
- [12] J. Jeong, E. Kwak, J.-h. Kim and K.-Y. Oh, "Multiphysics-informed DeepONet of a lithium-ion battery to predict thermal runaway," *PHM Society Asia-Pacific Conference*, 2023.
- [13] M. R. Niakan, S. S. Madani and M. Fowler, "Deep Learning-Based Battery Voltage Forecasting Using Current as Input: A Multi-Stage Approach for Time Series Prediction," *IEEE*, 2024.
- [14] K. Oka, N. Tanibata, M. Nakayama, S. Noguchi, M. Karasuyama, Y. Fujiware and T. Miyuki, "Deep learning based emulator for predicting voltage behaviour in lithium ion batteries," *Scientific Reports*, vol. 14, 2024.